

COP 3515 – Advanced Program Design

Semester Programming Project

Client Change Request (CCR-002)

Project: Student Information Management System (SIMS)

Customer: University Registrar's Office

Priority: High

Assigned To: Student Developer

Sprint: Week 2

Due Date: Beginning of Lesson 6

Business Background

The University Registrar's Office was pleased with the initial prototype of the Student Information Management System (SIMS). The software successfully records basic student information and displays a formatted summary.

The Registrar has identified the next enhancement needed for the system. Rather than recording only a student's GPA, the Registrar would like the software to record grades for the student's current courses. This will allow advisors to better evaluate academic performance before calculating future GPAs and academic standing.

Since every student enrolled this semester is taking five courses, the Registrar would like the software to record the grades for those five courses and provide a summary of the student's academic performance.

This enhancement should be added to the existing SIMS application developed during **CCR-001**. Do **not** create a new program. Continue building on your existing code.

Business Requirements

The Registrar would like the program to continue performing **all functionality from CCR-001**.

In addition, the software should:

1. Allow the user to enter the student's grades **for five courses**.
2. Store the grades in the program.
3. Calculate:
 - Average grade
 - Highest grade
 - Lowest grade
4. Display a formatted grade report.
5. Display both the student information from CCR-001 and the new grade summary.

At this stage, the system only needs to manage information **for one student**.

Acceptance Criteria

The project will be considered complete when all of the following requirements have been met.

- ☐ The program compiles without errors or warnings.
- ☐ All functionality from **CCR-001** continues to work correctly.
- ☐ The user can enter grades for five courses.
- ☐ The program stores all five grades correctly.
- ☐ The average grade is calculated correctly.
- ☐ The highest grade is identified correctly.
- ☐ The lowest grade is identified correctly.
- ☐ A neatly formatted grade report is displayed.
- ☐ Student information and grade information are displayed together.
- ☐ Output is professionally formatted and easy to read.

Technical Constraints

For this week's assignment, you may **only** use C language features covered in class.

Permitted concepts include:

- Variables
- Constants
- Enumerations (enum)
- Arrays
- Mathematical operators
- User input
- Formatted output
- Basic algorithms
- Selection statements introduced through this lesson

Do **not** use language features that have not yet been introduced in class.

Out of Scope

The Registrar is **not** requesting any of the following features at this time.

- Multiple students
- Functions
- Loops
- Files
- Menus
- Searching
- Editing grades
- Letter grade calculations
- Academic standing

- Dynamic memory allocation

These features will be requested in future Client Change Requests.

Questions for the Customer

Before writing any code, identify questions you would ask the customer to clarify the requirements.

Examples include:

1. Should grades be stored as integers or decimal values?
 2. What grading scale is being used (0–100, percentage, points, etc.)?
 3. Can grades contain fractional values such as 89.5?
 4. What should happen if an invalid grade is entered?
 5. Should the five courses have names, or only grades?
 6. Should the average be rounded to a specific number of decimal places?
 7. What should happen if two courses share the highest grade?
 8. What should happen if two courses share the lowest grade?
 9. Should the report identify which course earned the highest and lowest grades?
 10. Will future versions support more than five courses?
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Example Program Execution

Example 1

User Input

Student ID: 1234567

Student Name: Alice Johnson

Current GPA: 3.84

Course Grades

95

88

91

84

100

Expected Output

Student Information Management System

Version 2.0

Welcome to SIMS

Student Summary

Student ID : 1234567

Student Name : Alice Johnson

Current GPA : 3.84

Course Grades

Course 1 : 95

Course 2 : 88

Course 3 : 91

Course 4 : 84

Course 5 : 100

Average Grade : 91.60

Highest Grade : 100

Lowest Grade : 84

Example 2

User Input

Student ID: 1002003

Student Name: Michael Brown

Current GPA: 2.91

Course Grades

75

82

91

68

79

Expected Output

Average Grade : 79.00

Highest Grade : 91

Lowest Grade : 68

Regression Testing Requirements

Since this project is developed incrementally, **every previously implemented feature must continue to work correctly.**

Your submission must include evidence that you tested both the original functionality from **CCR-001** and the new functionality added in **CCR-002**.

At a minimum, include the following three test cases.

Test	Student ID	Student Name	GPA	Course Grades	Expected Average
1	1234567	Alice Johnson	3.84	95, 88, 91, 84, 100	91.60
2	1002003	Michael Brown	2.91	75, 82, 91, 68, 79	79.00
3	9876543	Christopher Williams	4.00	100, 100, 100, 100, 100	100.00

For **each** test case include:

- Input entered
 - Program output
 - Expected output
 - Pass/Fail result
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AI Usage Policy

Artificial Intelligence tools (such as ChatGPT, Copilot, Claude, or Gemini) may be used during this assignment.

However, you are responsible for all submitted work.

If AI-generated code contains errors, those errors are your responsibility.

Your grade is based on the correctness of your submitted program, **not** on how the code was produced.

Submission Requirements

Submit the following files to Canvas in a single ZIP file.

- main.c
- Any additional source files (if applicable)

Include a PDF containing:

- Screenshots of the required regression tests
 - A brief description of any bugs encountered
 - A description of how the bugs were corrected
 - Answers to the **Questions for the Customer** section
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Looking Ahead

This is the **second** release of the Student Information Management System (SIMS).

Future Client Change Requests will require you to continue modifying this same codebase. Upcoming enhancements will include decision making, functions, files, pointers, dynamic memory allocation, and other software engineering concepts.

Important: Professional software systems evolve over time. Do **not** start over each week. Continue improving your existing application while ensuring that previously implemented features continue to function correctly through regression testing.